

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

FINITE ELEMENT ANALYSIS

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) List the key steps involved in the Finite Element Method (FEM). | 2M |
| (b) Mention two demerits of the Finite Element Method (FEM). | 2M |
| (c) What are shape functions, and why are they important in finite element analysis? | 2M |
| (d) List the types of elements used for plane stress analysis. | 2M |
| (e) What is the role of an element stiffness matrix in finite element analysis? | 2M |
| (f) How does the size of the stiffness matrix differ between 3-node triangular and 4-node rectangular elements? | 2M |
| (g) What are the advantages of using Iso-parametric elements in 2D analysis? | 2M |
| (h) Differentiate between Lagrangian and Serendipity elements. | 2M |
| (i) Explain the trapezoidal rule used in numerical integration. | 2M |
| (j) Explain the term 'assembly of elements' in finite element analysis. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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| 2 | (a) Explain the principle of minimum potential energy and its significance in FEM. | 5M |
| | (b) Define the process of discretization in FEM and explain its importance in the analysis of complex structures. | 5M |
- OR**
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|---|--|----|
| 3 | (a) Briefly describe the Rayleigh-Ritz method and provide an example of its application in solving structural problems. | 5M |
| | (b) Write the general equilibrium equations for a 2D elastic body and explain their significance in structural analysis. | 5M |
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|---|--|----|
| 4 | (a) What is the natural coordinate system? How does it differ from the Cartesian coordinate system in finite element analysis? | 5M |
| | (b) Explain the requirements of convergence and compatibility in finite element analysis. | 5M |
- OR**
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|---|--|-----|
| 5 | Derive the shape functions for a two-noded 1D bar element. | 10M |
| 6 | Derive the stiffness matrix for a 3-node triangular element in a two-dimensional plane stress problem. | 10M |
- OR**
- | | | |
|---|---|-----|
| 7 | Explain the steps involved in calculating the element stiffness matrix for a 4-node rectangular element using bilinear shape functions. | 10M |
| 8 | Derive the strain-displacement relationship for the CST element. | 10M |
- OR**
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|----|---|-----|
| 9 | Derive the stiffness matrix for a 4-noded Axi-Symmetric Iso-Parametric element. | 10M |
| 10 | Derive the expression for the Simpson's 1/3rd rule for numerical integration. | 10M |
- OR**
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|----|---|----|
| 11 | (a) Explain the concept of static condensation in finite element analysis and its importance in reducing computational effort. | 5M |
| | (b) What are the advantages and limitations of using static condensation for solving large systems of equations in structural analysis? | 5M |
